

Matthieu Vinet

Final-year robotics engineering student · Polytech Sorbonne

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Final-year robotics internship · 6 months · From mid-February 2027

I enjoy building robots, working on mechanics, electronics and code. During my internship at Botshare.ai in Berlin, I brought a mobile robot back into operation and developed its vision-based docking system for wireless charging. My projects focus on control, simulation and my first experiments with reinforcement and imitation learning.

EXPERIENCE

Robotics engineering intern · Botshare.ai, Berlin

Apr. - Jul. 2026 · [Project page](#)

- Brought the robot back into operation: reworked its firmware using Linorobot and the hardware pinout, then enabled autonomous navigation with ROS 2 and Nav2.
- Built camera-based docking for wireless charging (AprilTag bundle, visual servoing), from Gazebo to the physical robot, made reliable despite glare, blur and people walking past. [Published a technical article](#).
- ROS 2 package openamrobot_docking; pull requests reviewed and merged on 3 repositories.

Robotech · Polytech Sorbonne robotics club

Since Sep. 2024

Helped restart the club and contributed to team projects; treasurer from April 2025 to April 2026.

ROBOTICS PROJECTS

SO-101 arms · Imitation learning

Robotech · 2026 · [Project page](#)

Printed and calibrated a leader/follower pair. Trained an ACT policy with LeRobot on 40 teleoperated demonstrations (grasping a tape roll): first successes, not yet robust.

Inverted-pendulum robot

Personal project · 2026 · [Project page](#)

Designed and built a balancing robot with 200 Hz cascade control on ESP32. Then trained a PPO policy in MuJoCo, ran it on the robot, and narrowed the sim-to-real gap by recalibrating the motor model.

Biped robot · Humanoid lower body

Robotech · Team of 3 · Ongoing

Led the design study of a biped with six joints per leg: simplified MuJoCo model, joint torques by inverse dynamics, actuator selection, budget. Next, once funded: CAD, build, then walking policies trained in simulation.

AGRI-BOT · Fourth-year industrial project

Polytech Sorbonne · 2026 · [Project page](#)

Monitoring robot for the Polytech smart greenhouse. Built the web teleoperation interface from scratch (live video, map-based missions, ROS 2 bridge) and the motorised pantograph arm that raises the camera from 30 to 140 cm.

Other projects (DIY brushless motor, RC plane, gesture-tracking camera, chess AI in C...) at matthieu-vinet.fr.

SKILLS

Robotics and control: ROS 2, Nav2, Gazebo, MuJoCo, system identification, inverse dynamics, OpenCV, AprilTags, visual servoing

Programming and embedded systems: Python, C/C++, ESP32, Git (code review, pull requests), Linux

Design and fabrication: SolidWorks, 3D printing, laser cutting

AI / ML, currently learning: PyTorch, imitation learning (ACT, LeRobot), RL (PPO, Stable-Baselines3), through ML coursework and personal projects.

EDUCATION

Engineering degree in Robotics · Polytech Sorbonne

2024 - 2027 · Expected 2027

Integrated preparatory programme · Polytech Tours

2022 - 2024

LANGUAGES

French: native · **English:** TOEIC Listening & Reading 780/990, taken before the Berlin internship.